

CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

Abinaya Sri J(192524429)

EXPERIMENT: PLANETS DATABASE

AIM

To write a Prolog program to create a database of planets and retrieve their properties using queries.

OBJECTIVE

1. To understand the concept of facts and rules in Prolog.
2. To create a simple Planet Database using Prolog.
3. To retrieve the properties of planets using queries.
4. To find planets based on their type, size, and temperature.

ALGORITHM

Step 1: Start.

Step 2: Define facts for each planet with its type, size, temperature, and position.

Step 3: Define the planet_properties predicate to retrieve the properties of a planet.

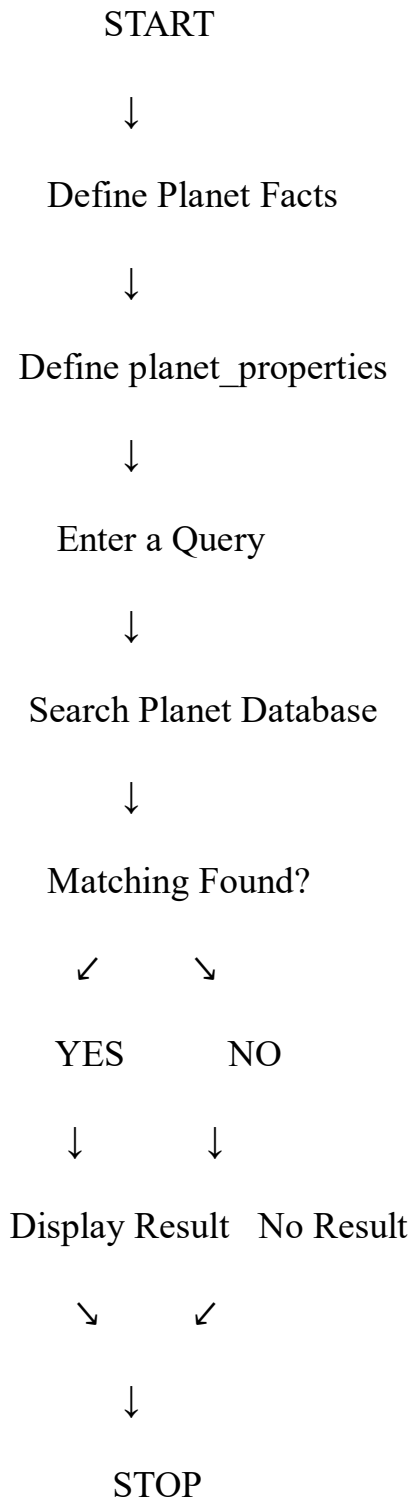
Step 4: Give a query to find the properties of a particular planet.

Step 5: Give a query to find planets based on their type, size, or temperature.

Step 6: Display the matching planets and their properties.

Step 7: Stop.

FLOWCHART



PROGRAM:

% define some facts about planets and their properties

planet(mercury, rocky, small, hot, closest_to_sun).

planet(venus, rocky, small, hot, 2nd_closest_to_sun).

planet(earth, rocky, medium, temperate, 3rd_closest_to_sun).

planet(mars, rocky, small, cold, 4th_closest_to_sun).

planet(jupiter, gas_giant, large, cold, 5th_closest_to_sun).

planet(saturn, gas_giant, large, cold, 6th_closest_to_sun).

planet(uranus, ice_giant, large, cold, 7th_closest_to_sun).

planet(neptune, ice_giant, large, cold, 8th_closest_to_sun).

% define a predicate to look up a planet's properties by name

planet_properties(Name, Type, Size, Temperature, Position) :-

planet(Name, Type, Size, Temperature, Position).

% example query: find the properties of earth

?- planet_properties(earth, Type, Size, Temperature, Position).

Type = rocky,

Size = medium,

Temperature = temperate,

Position = 3rd_closest_to_sun.% example query: find the planets that are gas giants

?- planet_properties(Name, gas_giant, _, _, _).

Name = jupiter ;

Name = saturn.

% example query: find the planets that are small and hot

66

?- planet_properties(Name, _, small, hot, _).

OUTPUT

```
?-  
% c:/Users/Abinaya Sri J/OneDrive/Desktop/Prolog exercise/PRG1.pl compiled 0.00 sec, 9 clauses  
?- planet_properties(earth, Type, Size, Temperature, Position).  
Type = rocky,  
Size = medium,  
Temperature = temperate,  
Position = third_closest_to_sun.  
  
?- planet_properties(Name, gas_giant, _, _, _).  
Name = jupiter ;  
Name = saturn.  
  
?- planet_properties(Name, _, small, hot, _).  
Name = mercury ;  
Name = venus ;  
false.
```

RESULT

Thus, the Prolog program for PLANETS DATABASE was successfully executed, and the required planet properties were retrieved using Prolog queries.